ALGEBRA 1 · LINEAR FUNCTIONS AND GRAPHING

Slope-Intercept Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Slope-Intercept Explorer** on Mr. K's Classroom: Algebra 1 → Linear Functions and Graphing → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. In y = mx + b, what do you think happens to the line when b gets bigger?

   The line moves up.

2. What do you think a line with a negative slope looks like?

   It goes down from left to right.

2 Explore use the tool and record what you see

Change b (keep rise 2, run 1, so m = 2)

b	Equation	y-intercept	Compared with y = 2x
–3	**y = 2x – 3**	**(0, –3)**	**down 3**
0	**y = 2x**	**(0, 0)**	**the same line**
2	**y = 2x + 2**	**(0, 2)**	**up 2**
5	**y = 2x + 5**	**(0, 5)**	**up 5**

Change the slope (keep b = 1)

Rise	Run	Slope m	Uphill, downhill or flat?	Steeper than y = x?
1	2	**$\frac{1}{2}$**	**uphill**	**no**
3	1	**3**	**uphill**	**yes**
–2	1	**–2**	**downhill**	**yes**
–1	3	**$-\frac{1}{3}$**	**downhill**	**no**
0	4	**0**	**flat**	**no**

Compare with

Turn on the parallel and perpendicular lines.

Your slope	Parallel slope	Perpendicular slope
$\frac{2}{3}$	**$\frac{2}{3}$**	**$-\frac{3}{2}$**
–4	**–4**	**$\frac{1}{4}$**
1	**1**	**–1**

3 Write the rule fill in each blank from what you observed

1. Changing b moves the line **up** or **down** without changing how steep it is.
2. When b is negative, the line crosses the y-axis **below** the origin.
3. When m is positive the line goes **up** from left to right. When m is negative it goes **down**.
4. The larger the absolute value of m, the **steeper** the line.
5. Parallel lines have **equal** slopes. Perpendicular slopes are opposite **reciprocals**.

4 Check no tool: show your work

1. Give the slope and y-intercept of $y = -3x + 4$.

   m = –3; (0, 4)

2. Write the equation of the line with slope $\frac{1}{2}$ and y-intercept $(0, -2)$.

   $y = \frac{1}{2}x - 2$

3. Write the line parallel to $y = 5x - 1$ that crosses the y-axis at 3.

   y = 5x + 3

4. What slope is perpendicular to $y = -\frac{2}{5}x + 7$?

   $\frac{5}{2}$

5 Reflect

Explain what the slope tells you, using the words rise and run.

Slope is $\frac{\text{rise}}{\text{run}}$: how much y changes for each 1-unit step to the right. Positive slopes rise, negative slopes fall.